DAY 19 · THEORY HANDOUT

Control a Local Model's Replies

Compare output limits, temperature, context space, and a reusable Modelfile.

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Laxmi Kant Tiwari · kgptalkie.com

The idea and its useful terms

Compare output limits, temperature, context space, and a reusable Modelfile.

Temperature

Changes how varied token choices can be.

Context window

Space for the input and generated tokens.

Modelfile

A text recipe for a base model and its settings.

A low temperature often gives more stable replies. It does not prove correctness or guarantee identical results on all systems.

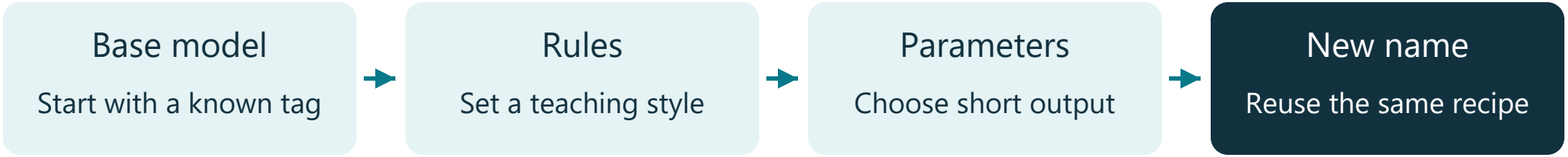
A reply limit caps generated tokens. If it is too small, the answer can stop mid-sentence. Context space is a different limit.

> A Modelfile changes a recipe and defaults. It does not teach new facts by training weights.

Follow the work, one step at a time

Each step has a job. We can inspect its output before moving on.

Read the path from left to right.



Each arrow means: pass the result to the next step.

temperature changes token selection. num_predict limits newly generated tokens. num_ctx sets context space for the model run. They control different things, so change one setting at a time.

A Modelfile starts with FROM to name the base model. SYSTEM sets its default job, and PARAMETER sets defaults. Creating a new tag from the recipe reuses the base weights; it is not a training run.

> If the answer is wrong, inspect the input and the intermediate results. The last sentence alone cannot show which step failed.

A small example we can check

The values here are classroom examples. The notebook shows actual results.

temperature 0

Useful for extraction.

Check facts anyway.

temperature 0.8

Try a poem or idea list.

Answers may vary more.

num_predict 128

Cap reply length.

Not a word-count rule.

Read the given input first. Point to the fact or value needed for the requested output.

Run the matching example, then try the fresh input at the end of the notebook. Compare both results with the known input.

> A Modelfile changes a recipe and defaults. It does not teach new facts by training weights.

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Choose the setting or tool for its job

A different choice changes the work. These are the choices we compare.

SYSTEM

Set the assistant's job.

Keep changing data outside.

PARAMETER

Choose default settings.

Test them with real inputs.

FROM

Choose the base model.

This is not fine-tuning.

Run the same short prompt with two temperatures. Compare the saved replies.

Lower the token limit and notice whether the answer is cut short.

> Read the supplied Modelfile and change only the teaching rule. Create a new model name in the terminal if you want to try it.

Practice, check, and read the full lesson

Use the notebook to try the steps. Use the article when you need the longer explanation.

1. Build

Run the small example.

Inspect each step.

2. Change

Try the fresh input.

Keep the goal clear.

3. Check

Compare with the source.

Keep mistakes visible.

Read the supplied Modelfile and change only the teaching rule. Create a new model name in the terminal if you want to try it.

Read the complete tutorial breakdown for this section in our KGP Talkie article.

> A Modelfile changes a recipe and defaults. It does not teach new facts by training weights.

DAY 19 · COMPLETE

Thank You

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